Bipartite Ranking with the Pairwise Hinge is Balanced Classification over a Transportation Polytope

Jesse Glass

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Abstract

We show that the dual of the pairwise-hinge bipartite ranking SVM (RankSVM) is *exactly* the dual of a balanced cost-sensitive classification SVM, restricted to a capacitated transportation polytope. Under a margin-2 convention the two dual objectives coincide – not merely up to constants – and the classification intercept constraint arises automatically. We characterize the restricted domain exactly: it is the convex hull of per-example *misordering-count* vectors, described by Gale-type inequalities, with membership decidable by a single maximum-flow computation that also returns a dual witness. Consequences: (i) the balanced solution lower-bounds the RankSVM dual optimum, and any transportation-feasible point upper-bounds it, giving a computable optimality certificate; (ii) every Frank–Wolfe step on the pairwise dual is a sample-weighted classification problem whose weights count misordered opposite-class examples, and balanced classification is precisely the first Frank–Wolfe step; (iii) at the gradient level the equivalence holds for *any* differentiable scorer: the parameter gradient of a decomposable pairwise loss equals the gradient of a count-weighted univariate loss, whose weights for the pairwise hinge are computable in $O(N \log N)$ without forming pairs. This geometry unifies a family of known efficiency results and reweighting heuristics – structural-SVM cutting planes, sorted-gradient RankSVM solvers, LambdaMART's per-document lambdas, hard-example mining – as views of one object. Experiments confirm the theory's predictions, including the negative ones: count-reweighted classifiers match RankSVM and pairwise-loss networks everywhere at a fraction of the cost, while offering only situational gains over static balanced weighting, exactly as AUC-consistency theory implies.

1 Introduction

Bipartite ranking asks for a scoring function that places positive examples above negative ones; its canonical empirical objective is the area under the ROC curve (AUC), and its canonical convex surrogate is the pairwise hinge over all positive–negative pairs (Herbrich et al., 2000; Joachims, 2002). Taken literally, the pairwise objective couples n^+n^- terms, and a substantial literature exists on evaluating or optimizing it without enumerating pairs (Joachims, 2005; Chapelle and Keerthi, 2010; Airola et al., 2011; Ying et al., 2016). A parallel statistical literature shows that pairs are unnecessary in a deeper sense: minimizing a class-balanced *univariate* loss is AUC-consistent, with explicit surrogate regret bounds (Kotłowski et al., 2011; Agarwal, 2014).

This paper gives the exact finite-sample geometry underlying both observations. Our main results concern the Lagrangian duals of the two problems. Let the pairwise SVM be taken with margin 2 (a rescaling of the standard formulation; Remark 1) and let the balanced classification

SVM weight each example by the empirical frequency of the *opposite* class. Then, under the natural row/column-sum map from pair multipliers to example multipliers:

1. **Equivalence (Theorem 1).** The pairwise dual objective *equals* the balanced classification dual objective – same quadratic form, same linear term – and the intercept constraint of the classification dual holds automatically. The pairwise dual is the balanced dual restricted to a subdomain $\mathcal{T}$.

2. **Geometry (Theorem 2).** $\mathcal{T}$ is a capacitated transportation polytope: the convex hull of scaled *misordering-count* vectors, cut out by Gale-type inequalities (Gale, 1957), with membership decidable by one max-flow computation on a network with $n^+ + n^- + 2$ nodes (Corollary 1). It is strictly smaller than the balanced dual's box-plus-hyperplane domain.

3. **Certificate (Corollary 2).** The balanced optimum lower-bounds the pairwise optimum; any point of $\mathcal{T}$ upper-bounds it. The gap is computable and vanishes exactly when the balanced solution is transportation-feasible.

4. **Beyond linear models (Proposition 1).** For any differentiable scorer, the parameter gradient of a decomposable pairwise loss equals the gradient of a per-example loss weighted by (soft) violation counts, frozen at the current scores. Count-reweighted training *is* pairwise training at the gradient level, at $O(N \log N)$ per evaluation for the hinge.

The picture that emerges is summarized by Corollary 3: running Frank–Wolfe (Frank and Wolfe, 1956; Jaggi, 2013) on the pairwise dual is running Frank–Wolfe on $\mathcal{T}$, each linear-minimization step is a weighted classification problem whose weights are misordering counts, and *balanced classification is exactly the first step.* This explains, in one frame, why balanced univariate training is so hard to beat for AUC (Kotłowski et al., 2011), why structural-SVM cutting planes for ROC area reduce to sorting (Joachims, 2005), why primal RankSVM gradients collapse to per-example counts (Chapelle and Keerthi, 2010), why gradient-boosted ranking works from per-document "lambdas" (Burges, 2010), and why hard-example mining heuristics (Shrivastava et al., 2016; Lin et al., 2017) succeed: each example's principled "hardness" weight is the number of opposite-class examples that outrank it.

We claim no new state-of-the-art method, and our experiments (Section 6) are designed to test the theory's predictions rather than to win benchmarks. They confirm: count-reweighted classifiers reproduce RankSVM (linear) and pairwise-logistic training (MLPs) within noise on every dataset tried, at 1.3–4.4× lower cost per epoch and better asymptotic complexity; iterated reweighting with cross-fitted model selection behaves as a "free option" over balanced weighting – it selects the balanced solution when that is optimal and harvests modest gains when headroom exists; and, as consistency theory predicts, neither pairwise training nor its reweighted equivalent broadly beats plain or balanced univariate baselines on generic tabular data. The contribution is the geometry, the certificate, and the unification.

2 Related Work

Efficient pairwise training. That the pairwise hinge can be optimized without materializing pairs is well established. Joachims (2005) trains an ROC-area structural SVM whose cutting-plane oracle runs in $O(N \log N)$ by sorting; Joachims (2006) shows the error-rate structural SVM is equivalent to a classification SVM. Chapelle and Keerthi (2010) solve primal

RankSVM with Newton methods whose objective and gradient are evaluated in $O(N \log N)$; Airola et al. (2011) achieve linearithmic training with order-statistic trees. In learning to rank, LambdaRank/LambdaMART collapse pairwise gradients into per-document weights (Burges, 2010). For the square surrogate, Ying et al. (2016) remove pairs entirely via a min-max reformulation, the basis of large-scale deep AUC maximization (Yuan et al., 2021; Yang and Ying, 2022). Our contribution relative to this line is not efficiency but structure: these algorithms implicitly traverse the transportation polytope $\mathcal{T}$; we name the object, characterize it exactly, and equip it with a membership test and an optimality certificate none of the above provide.

Statistical equivalences. Kotłowski et al. (2011) and Agarwal (2014) show balanced univariate losses are AUC-consistent with explicit regret transfer; Ertekin and Rudin (2011) prove AdaBoost and RankBoost are equivalent in a precise sense; Clémençon et al. (2008) give the U-statistics theory of pairwise ERM. These are asymptotic or population-level statements; Theorem 1 is their exact finite-sample, fixed-regularization counterpart at the level of dual programs, and Corollary 2 quantifies the finite-sample gap between balanced and pairwise solutions.

Frank–Wolfe for structured duals. Block-coordinate Frank–Wolfe for structural SVMs (Lacoste-Julien et al., 2013) popularized dual optimization via linear-minimization oracles. Corollary 3 identifies the pairwise-ranking LMO with a weighted classification problem, which makes plug-in, dual-free implementations possible (Section 5).

Reweighting heuristics. Online hard example mining (Shrivastava et al., 2016), focal loss (Lin et al., 2017), and margin-based weighted-classification losses in metric learning (Deng et al., 2019) weight examples by difficulty. Proposition 1 supplies the principled special case: for bipartite objectives, the exact pairwise-loss gradient is recovered by weighting each example with its count of violated pairs.

3 Dual Coefficient Mappings

This section establishes the precise relationship between the pairwise (bipartite-ranking) SVM and the balanced cost-sensitive classification SVM at the level of their dual programs. The main results are: (i) under the natural row/column-sum map, the pairwise dual *objective* coincides exactly with the balanced classification dual objective (Theorem 1); (ii) the pairwise dual feasible set maps onto a *capacitated transportation polytope* strictly contained in the balanced dual feasible set, with the intercept constraint of the classification SVM arising automatically (Theorems 1 and 2); and (iii) membership in this polytope is testable in polynomial time (Corollary 1). Consequently, bipartite ranking with the pairwise hinge *is* balanced cost-sensitive classification with additional transportation constraints on the dual variables, and each Frank–Wolfe vertex of the pairwise dual corresponds to a sample-weighted classification problem whose weights count misordered opposite-class examples (Corollary 3). These results license the dual-variable-free algorithms of the next section: we may optimize the pairwise dual entirely in the space of univariate Lagrangian coefficients, successively tightening the balanced domain toward the transportation polytope, and stopping early whenever leaving the polytope improves the margin.

3.1 Setup and conventions

Let x_i, $i = 1, \dots, n^+$ denote the positive examples and x_j, $j = 1, \dots, n^-$ the negative examples, with $N = n^+ + n^-$, $p = n^+/N$, $q = n^-/N$, and labels $t_k \in \{+1, -1\}$ indexed by $k = 1, \dots, N$ (index i ranges over positives, j over negatives, k over all examples, and $\pi = (i, j)$ over positive–negative pairs). Write $z_\pi = x_i - x_j$ for the pair difference vectors, $Q = tt^\top \odot XX^\top$ for the standard SVM Gram operator, and

$$\bar{C} \;=\; \frac{C}{N^2} \tag{1}$$

for the per-pair regularization weight.

Pairwise hinge, margin-2 convention. The pairwise (RankSVM) primal is taken with margin 2:

$$\min_{w} \;\; \frac{1}{2}\|w\|^2 \;+\; \bar{C}\sum_{i=1}^{n^+}\sum_{j=1}^{n^-}\big[\, 2 - \langle w, x_i - x_j\rangle \,\big]_+. \tag{2}$$

Its Lagrangian dual, with one multiplier $\rho_\pi = \rho_{ij}$ per pair, is

$$\min_{\rho} \;\; \frac{1}{2}\rho^\top \widetilde{Q}\rho \;-\; 2\,\mathbb{1}^\top\rho \qquad \text{s.t.} \quad 0 \le \rho_\pi \le \bar{C} \;\; \forall\pi, \tag{3}$$

where $\widetilde{Q}_{\pi\pi'} = \langle z_\pi, z_{\pi'}\rangle$ and $w = \sum_\pi \rho_\pi z_\pi$ at optimality. There is no equality constraint because the intercept cancels within each pair.

Remark 1 (Margin 2 is standard RankSVM, rescaled). For any w, substituting $v = w/2$ in (2) gives

$$\frac{1}{2}\|w\|^2 + \bar{C}\sum_{\pi}\big[2 - \langle w, z_\pi\rangle\big]_+ = 4\left(\frac{1}{2}\|v\|^2 + \frac{\bar{C}}{2}\sum_{\pi}\big[1 - \langle v, z_\pi\rangle\big]_+\right).$$

Hence the margin-2 problem with parameter C has the same minimizer, up to the scaling $w = 2v$, as the unit-margin RankSVM with parameter $C/2$. The induced ranking function, the AUC, and the regularization path are unchanged; only the parameterization of C shifts. The margin-2 convention is the natural one here because it is the margin produced by summing the two univariate hinges of a pair: $[1 - \langle w, x_i\rangle]_+ + [1 + \langle w, x_j\rangle]_+ \ge [2 - \langle w, x_i - x_j\rangle]_+$.

Balanced classification. The balanced (class-weighted) binary SVM with intercept weights each positive example by $\bar{C}n^-$ and each negative example by $\bar{C}n^+$:

$$\min_{w,b} \;\; \frac{1}{2}\|w\|^2 \;+\; \bar{C}\,n^-\sum_{i=1}^{n^+}\big[1 - (\langle w, x_i\rangle + b)\big]_+ \;+\; \bar{C}\,n^+\sum_{j=1}^{n^-}\big[1 + (\langle w, x_j\rangle + b)\big]_+. \tag{4}$$

(Equivalently, per-class weights $\frac{C}{N}q$ and $\frac{C}{N}p$: each example is weighted by the empirical frequency of the *opposite* class.) Its dual, with one multiplier λ_k per example, is

$$\min_{\lambda} \;\; D(\lambda) \;:=\; \frac{1}{2}\lambda^\top Q\lambda \;-\; \mathbb{1}^\top\lambda \qquad \text{s.t.} \quad \lambda \in \mathcal{B}, \tag{5}$$

$$\mathcal{B} \;=\; \Big\{\lambda \in \mathbb{R}^N_{\ge 0} \;:\; \lambda_i \le \bar{C}n^- \;\forall i,\;\; \lambda_j \le \bar{C}n^+ \;\forall j,\;\; \textstyle\sum_k t_k\lambda_k = 0\Big\}, \tag{6}$$

with $w = \sum_k t_k\lambda_k x_k$ at optimality. The equality constraint in (6) is the usual consequence of the intercept b.

3.2 Background: Frank–Wolfe and Lagrangian dual coefficients

The structural SVM and its (block-coordinate) Frank–Wolfe optimizers (Joachims, 2005, 2006; Lacoste-Julien et al., 2013) work with dual coefficients α indexed by *joint labelings* (vertices) rather than by examples. For a decomposable hinge, each vertex $s \in \{0,1\}^N$ marks which examples are margin-violated, and the linear operator $\mathcal{S}$, whose columns are the vertices, maps vertex coefficients to example coefficients:

$$\lambda \;=\; \mathcal{S}\alpha, \qquad \alpha \in \Delta_C := \Big\{\alpha \geq 0 : \textstyle\sum_s \alpha_s = C\Big\}. \tag{7}$$

The two background lemmas below record that nothing is lost by tracking λ instead of α; they are essentially the known equivalence between the error-rate structural SVM and the classification SVM (Joachims, 2006, Theorem 1) and are stated in the form used later.

Lemma 1 (Per-example blocks: bijection)**.** *In the block-separable formulation (one block per example, block mass C_k equal to that example's primal weight), each block has two vertices – violated and satisfied – with coefficients $\alpha_{k1}+\alpha_{k0} = C_k$. The map $\lambda_k = \alpha_{k1}$ is a bijection between the per-block Frank–Wolfe domain and the Lagrangian box $\lambda_k \in [0, C_k]$, applied independently across blocks.*

Proof. Immediate: $\alpha_{k1} \in [0, C_k]$ determines $\alpha_{k0} = C_k - \alpha_{k1}$ uniquely, and every $\lambda_k \in [0, C_k]$ is attained. ☐

Lemma 2 (Holistic formulation: surjection onto the box)**.** *Let $\mathcal{S} \in \{0,1\}^{N\times 2^N}$ contain all binary vertices as columns. The map $\alpha \mapsto \mathcal{S}\alpha$ sends Δ_C onto $[0,C]^N$, using at most N vertices with nested supports; it is not injective for $N \geq 2$.*

Proof. (*Surjectivity: layer-cake construction.*) Fix $\lambda \in [0,C]^N$ and let $0 < v_1 < v_2 < \cdots < v_r$ be its distinct positive values, with $v_0 = 0$. Define nested vertices $s_m = \mathbb{1}\{\lambda \geq v_m\}$ and coefficients $\alpha_{s_m} = v_m - v_{m-1} > 0$ for $m = 1,\ldots,r$. For each coordinate k, $\sum_m \alpha_{s_m}(s_m)_k = \sum_{m:v_m\leq\lambda_k}(v_m - v_{m-1}) = \lambda_k$, so $\mathcal{S}\alpha = \lambda$. The coefficients sum to $v_r = \max_k \lambda_k \leq C$; assign the remainder $C - v_r$ to the zero vertex $s_0 = \mathbf{0}$. This uses $r \leq N$ nonzero vertices plus the zero vertex, all with nested supports.

(*Non-injectivity.*) For $N = 2$, the distinct points $\alpha = \frac{C}{2}\big(e_{(1,1)} + e_{(0,0)}\big)$ and $\widetilde{\alpha} = \frac{C}{2}\big(e_{(1,0)} + e_{(0,1)}\big)$ both map to $\lambda = (\frac{C}{2}, \frac{C}{2})$. ☐

For the pairwise loss, the vertices are subsets s of positive–negative pairs, and the corresponding column of $\mathcal{S}$ records, for each example, *how many* pairs in s it participates in:

$$(\mathcal{S}e_s)_i \;=\; \#\{\, j : (i,j) \in s \,\}, \qquad (\mathcal{S}e_s)_j \;=\; \#\{\, i : (i,j) \in s \,\}. \tag{8}$$

Under the margin-2 convention the structural loss of vertex s is $\Delta_s = 2|s|$ (each misordered pair contributes its margin), and since each pair touches exactly two examples, $\mathbb{1}^\top \mathcal{S} e_s = 2|s| = \Delta_s$. Summing over vertices:

$$\Delta^\top \alpha \;=\; \mathbb{1}^\top \mathcal{S}\alpha \;=\; \mathbb{1}^\top \lambda. \tag{9}$$

This is the pairwise analogue of the classification identity and is the bookkeeping that makes Theorem 1 exact. (Under a unit-margin convention the two sides differ by a factor of 2; see Remark 1.)

3.3 The pairwise dual is a constrained balanced dual

Define the row/column-sum map Φ from pair coefficients to example coefficients:

$$\Phi(\rho) \;=\; \lambda, \qquad \lambda_i = \sum_{j=1}^{n^-} \rho_{ij}, \qquad \lambda_j = \sum_{i=1}^{n^+} \rho_{ij}. \tag{10}$$

Theorem 1 (Dual equivalence)**.** *Let $\mathcal{R} = [0, \bar{C}]^{n^+ \times n^-}$ be the pairwise dual feasible set (3) and let $\lambda = \Phi(\rho)$. Then:*

(i) (Weight vectors agree.) $\sum_\pi \rho_\pi z_\pi = \sum_k t_k \lambda_k x_k$.

(ii) (Objectives agree.) $\frac{1}{2}\rho^\top \widetilde{Q}\rho - 2\,\mathbb{1}^\top \rho = \frac{1}{2}\lambda^\top Q\lambda - \mathbb{1}^\top \lambda = D(\lambda)$.

(iii) (Feasibility is preserved, intercept included.) $\Phi(\mathcal{R}) \subseteq \mathcal{B}$, where $\mathcal{B}$ is the balanced feasible set (6). In particular the equality constraint $\sum_k t_k \lambda_k = 0$ holds automatically for every $\lambda \in \Phi(\mathcal{R})$.

Consequently, with $\mathcal{T} := \Phi(\mathcal{R})$,

$$\min_{\rho \in \mathcal{R}} \frac{1}{2}\rho^\top \widetilde{Q}\rho - 2\,\mathbb{1}^\top \rho \;=\; \min_{\lambda \in \mathcal{T}} D(\lambda), \tag{11}$$

i.e. the pairwise dual is exactly the balanced classification dual restricted to the subdomain $\mathcal{T} \subseteq \mathcal{B}$.

Proof. (i) Splitting each pair difference,

$$\sum_\pi \rho_\pi z_\pi = \sum_i \sum_j \rho_{ij}(x_i - x_j) = \sum_i \Big(\sum_j \rho_{ij}\Big) x_i - \sum_j \Big(\sum_i \rho_{ij}\Big) x_j = \sum_i \lambda_i x_i - \sum_j \lambda_j x_j = \sum_k t_k \lambda_k x_k.$$

(ii) The quadratic terms both equal $\|w\|^2$ for the common w of part (i): $\rho^\top \widetilde{Q}\rho = \|\sum_\pi \rho_\pi z_\pi\|^2 = \|\sum_k t_k \lambda_k x_k\|^2 = \lambda^\top Q \lambda$. For the linear terms, each unit of ρ_{ij} contributes once to λ_i and once to λ_j, so

$$\mathbb{1}^\top \lambda = \sum_i \lambda_i + \sum_j \lambda_j = 2\sum_\pi \rho_\pi = 2\,\mathbb{1}^\top \rho.$$

(iii) Row and column sums of a matrix with entries in $[0, \bar{C}]$ obey $\lambda_i = \sum_j \rho_{ij} \le \bar{C} n^-$ and $\lambda_j = \sum_i \rho_{ij} \le \bar{C} n^+$, matching the balanced box in (6). Moreover $\sum_i \lambda_i = \sum_{ij} \rho_{ij} = \sum_j \lambda_j$, i.e. $\sum_k t_k \lambda_k = 0$: the total mass on positives equals the total mass on negatives because every pair contributes equally to both sides. Thus $\Phi(\rho) \in \mathcal{B}$.

Finally, (11) follows because by (ii) the objective is constant on the fibers of Φ, so minimizing over $\mathcal{R}$ and minimizing D over the image $\mathcal{T}$ are the same problem. □

Remark 2 (Φ is not injective, and this does not matter). For $n^+, n^- \ge 2$ the map Φ cannot be injective: it maps $n^+ n^-$ dimensions to $n^+ + n^-$ dimensions, and explicitly, the *rectangle move* replacing $(\rho_{ij}, \rho_{i'j'}, \rho_{ij'}, \rho_{i'j})$ by $(\rho_{ij} + \varepsilon, \rho_{i'j'} + \varepsilon, \rho_{ij'} - \varepsilon, \rho_{i'j} - \varepsilon)$ preserves every row and column sum. However, by Theorem 1(ii) the dual objective depends on ρ only through $\lambda = \Phi(\rho)$, so the reduced problem (11) loses nothing: any minimizer $\lambda^\star$ of D over $\mathcal{T}$ pulls back to a (non-unique) optimal ρ, and the primal solution w is recovered from $\lambda^\star$ alone.

3.4 The image is a capacitated transportation polytope

Theorem 1 reduces the pairwise dual to minimizing the balanced dual objective over $\mathcal{T} = \Phi(\mathcal{R})$. We now characterize $\mathcal{T}$ exactly. Write $\Lambda := \sum_i \lambda_i = \sum_j \lambda_j$ for the common one-sided mass of a point satisfying the equality constraint, and let $L^+(a)$ denote the sum of the a largest coefficients among the positive examples and $L^-(b)$ the sum of the b largest coefficients among the negative examples.

Theorem 2 (Exact image)**.** *$\mathcal{T}$ admits the following two descriptions.*

(i) (Vertex description.) $\mathcal{T}$ is the convex hull of the scaled misordering-count vectors:

$$\mathcal{T} \;=\; \operatorname{conv}\big\{\, \bar{C}\,\mathcal{S}e_s \;:\; s \subseteq \{1,\dots,n^+\} \times \{1,\dots,n^-\} \,\big\},$$

with $\mathcal{S}e_s$ the per-example pair counts of (8).

(ii) (Inequality description.) $\lambda \in \mathcal{T}$ if and only if $\lambda \geq 0$, $\sum_i \lambda_i = \sum_j \lambda_j = \Lambda$, and

$$L^+(a) + L^-(b) \;\leq\; \Lambda + \bar{C}\,ab \qquad \textit{for all } 1 \leq a \leq n^+,\ 1 \leq b \leq n^-. \tag{12}$$

Moreover $\mathcal{T} \subsetneq \mathcal{B}$ whenever $n^+, n^- \geq 2$: the box and equality constraints of $\mathcal{B}$ do not imply (12).

Proof. (i) The feasible set $\mathcal{R} = [0, \bar{C}]^{n^+ \times n^-}$ is a polytope whose vertices are $\bar{C}\,\mathbb{1}_s$ for pair subsets s (entries equal to $\bar{C}$ on s, zero elsewhere). The image of a polytope under a linear map is the convex hull of the images of its vertices, and $\Phi(\bar{C}\,\mathbb{1}_s) = \bar{C}\,\mathcal{S}e_s$ by (8).

(ii) Necessity of $\lambda \geq 0$ and of the equal-sum condition is Theorem 1(iii), so only (12) requires an argument. The question is feasibility of a transportation problem: does there exist $\rho \in [0, \bar{C}]^{n^+ \times n^-}$ with prescribed row sums λ_i and column sums λ_j? Model it as a flow network: source $\sigma \to$ row i with capacity λ_i; row $i \to$ column j with capacity $\bar{C}$; column $j \to$ sink τ with capacity λ_j. Feasibility is equivalent to a maximum flow of value Λ, which by max-flow/min-cut holds iff every σ–τ cut has capacity at least Λ. A cut is determined by the set S of rows and the set T' of columns on the source side; writing T for the complement of T', its capacity is $\sum_{i \notin S} \lambda_i + \bar{C}\,|S|\,|T| + \sum_{j \in T'} \lambda_j$. Requiring this to be at least $\Lambda = \sum_i \lambda_i$ yields

$$\sum_{i \in S} \lambda_i \;\leq\; \bar{C}\,|S|\,|T| + \sum_{j \notin T} \lambda_j \qquad \forall\, S, T.$$

For fixed cardinalities $a = |S|$, $b = |T|$, the left side is maximized by the a largest λ_i and the right side minimized by placing the b largest λ_j inside T; substituting $\sum_{j \notin T} \lambda_j = \Lambda - L^-(b)$ gives (12). Cases $a = 0$ or $b = 0$ are implied by nonnegativity and the equal-sum condition.

(*Strictness.*) Example 1 below exhibits, for $n^+ = n^- = 2$, a point of $\mathcal{B}$ violating (12); for general $n^+, n^- \geq 2$, pad it with zeros ($\lambda^+ = (2\bar{C}, 0, \dots, 0)$, $\lambda^- = (2\bar{C}, 0, \dots, 0)$): the box bounds $2\bar{C} \leq \bar{C} n^{\mp}$ still hold, and $a = b = 1$ still gives $4\bar{C} > \Lambda + \bar{C} = 3\bar{C}$. □

Example 1 ($\mathcal{T}$ is strictly smaller than box $\cap$ hyperplane). Let $n^+ = n^- = 2$ and consider $\lambda^+ = (2\bar{C},\, 0)$, $\lambda^- = (2\bar{C},\, 0)$. All box constraints of $\mathcal{B}$ hold ($2\bar{C} = \bar{C} n^-$) and the sums are equal, so $\lambda \in \mathcal{B}$. But with $a = b = 1$, $L^+(1) + L^-(1) = 4\bar{C} > \Lambda + \bar{C} = 3\bar{C}$, violating (12): the single positive example carrying mass $2\bar{C}$ would have to route more than the cell capacity $\bar{C}$ through the single negative example carrying mass. Concretely, no $\rho \in [0, \bar{C}]^{2 \times 2}$ has row sums $(2\bar{C}, 0)$ and column sums $(2\bar{C}, 0)$.

Corollary 1 (Polynomial-time membership and projection oracles)**.** *Whether $\lambda \in \mathcal{T}$ can be decided by a single maximum-flow computation on the bipartite network of Theorem 2(ii) (with $n^+ + n^- + 2$ nodes), or by checking the $n^+ n^-$ sorted prefix conditions (12) after two sorts (together with nonnegativity and the equal-sum check). A witness ρ with $\Phi(\rho) = \lambda$ is recovered from the flow itself. Consequently, the successive-tightening algorithms of Section 5 have an efficient certificate for "the current iterate is (nearly) pairwise-feasible," and Euclidean projection onto $\mathcal{T}$ is a quadratic program over the same flow polytope.*

3.5 Consequences

Corollary 2 (Sandwich certificate for the pairwise optimum)**.** *Let $\lambda^\star_{\text{bal}} \in \arg\min_{\lambda\in\mathcal{B}} D(\lambda)$ and $\lambda^\star_{\text{pair}} \in \arg\min_{\lambda\in\mathcal{T}} D(\lambda)$. Then for every $\lambda \in \mathcal{T}$,*

$$D(\lambda^\star_{\text{bal}}) \;\le\; D(\lambda^\star_{\text{pair}}) \;\le\; D(\lambda), \tag{13}$$

so any pairwise-feasible iterate $\lambda \in \mathcal{T}$ certifies its own suboptimality on the pairwise problem: $D(\lambda) - D(\lambda^\star_{\text{pair}}) \le D(\lambda) - D(\lambda^\star_{\text{bal}})$. In particular, if the balanced solution happens to satisfy $\lambda^\star_{\text{bal}} \in \mathcal{T}$, it is a pairwise (RankSVM) dual solution.

Proof. Immediate from $\mathcal{T} \subseteq \mathcal{B}$ (Theorem 1(iii)) and the shared objective D (Theorem 1(ii)). □

Corollary 3 (Frank–Wolfe vertices are weighted classification problems)**.** *Each Frank–Wolfe vertex s of the pairwise dual corresponds under $\lambda_s = \bar{C}\,\mathcal{S}e_s$ to the vector of misordering counts: example i receives weight proportional to the number of negatives in s paired above it, and example j to the number of positives paired below it. By Theorem 2(i), $\lambda_s \in \mathcal{T}$, and $\mathcal{T}$ is exactly the convex hull of these vertices; by (9) the Frank–Wolfe linearization $\langle \nabla, \alpha\rangle$-objective transfers to λ-space with the same value. Hence running Frank–Wolfe on the pairwise dual is running Frank–Wolfe on $\min_{\lambda\in\mathcal{T}} D(\lambda)$, and each linear-minimization step selects the count-weighted univariate problem used by the algorithms of the next section. At the first iterate ($w = 0$) every pair is violated, so the counts are n^- for every positive and n^+ for every negative: the balanced classification problem (4) is precisely the first Frank–Wolfe subproblem.*

Remark 3 (Relaxed trust regions). The trust-region subproblems of Section 5 optimize D over boxes $[0, \bar{C}\omega]$ with $\omega = \mathcal{S}e_s$ the current counts. The vertex $\bar{C}\omega$ itself lies in $\mathcal{T}$, but the box is *not* contained in $\mathcal{T}$ (coordinate-wise shrinking breaks the equal-mass constraint, and (12) may fail). The subproblems are therefore relaxations: their solutions can leave $\mathcal{T}$, and by Corollary 2 they lower-bound the pairwise optimum whenever they leave it. The algorithms exploit this deliberately – an iterate is pulled back toward $\mathcal{T}$ only when doing so does not worsen the margin – and Corollary 1 supplies the test.

4 Beyond Linear Models: the Gradient Identity

The dual equivalence above is specific to the SVM. Its engine, however, is a chain-rule fact that holds for any differentiable scorer, including deep networks.

Proposition 1 (Count-weighted gradients)**.** *Let $s_\theta : \mathbb{R}^d \to \mathbb{R}$ be differentiable in θ, write $s_k = s_\theta(x_k)$, and let $\ell : \mathbb{R} \to \mathbb{R}$ be a differentiable (or almost everywhere differentiable) pairwise*

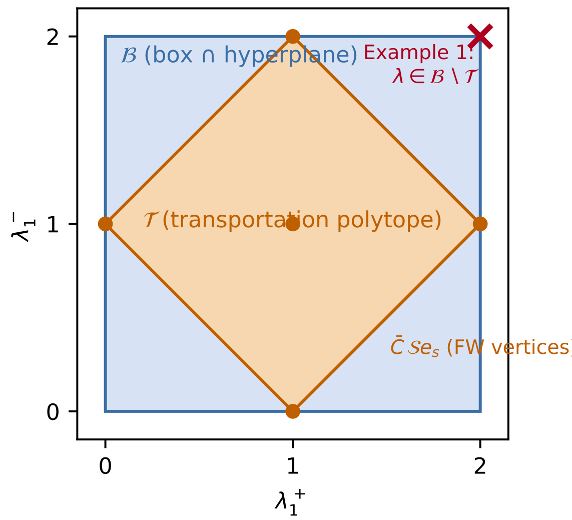

Figure 1: An exact two-dimensional slice of the dual domains for $n^+ = n^- = 2$, $\bar{C} = 1$, at one-sided mass $\Lambda = 2$: coordinates $(\lambda_1^+, \lambda_1^-)$ with $\lambda_2^+ = 2 - \lambda_1^+$, $\lambda_2^- = 2 - \lambda_1^-$. The balanced feasible set $\mathcal{B}$ (box ∩ intercept hyperplane) slices to the square; the transportation polytope $\mathcal{T}$ slices to the inscribed diamond – here (12) reduces to $|\lambda_1^+ - 1| + |\lambda_1^- - 1| \le 1$ – whose corners and center are the misordering-count vertices $\bar{C}\,\mathcal{S}e_s$ lying in the slice. The cross marks the point of Example 1, feasible for $\mathcal{B}$ but not for $\mathcal{T}$. The dual objective D is identical on $\mathcal{T}$ for both problems (Theorem 1).

surrogate applied to score differences,

$$L_{\text{pair}}(\theta) = \sum_{i=1}^{n^+} \sum_{j=1}^{n^-} \ell\big(s_i - s_j\big).$$

Define per-example weights frozen at the current scores,

$$c_i = -\sum_j \ell'(s_i - s_j), \qquad c_j = -\sum_i \ell'(s_i - s_j).$$

Then

$$\nabla_\theta L_{\text{pair}} = \nabla_\theta \Big(\sum_j c_j\, s_j - \sum_i c_i\, s_i \Big)\Big|_{c\ \text{frozen}}.$$

For the margin hinge $\ell(u) = [\,m - u\,]_+$ the weights are the violation counts $c_i = \#\{j : s_i - s_j < m\}$ and $c_j = \#\{i : s_i - s_j < m\}$, computable for all examples in $O(N \log N)$ by two sorts. For the logistic surrogate $\ell(u) = \log(1 + e^{-u})$ they are the soft counts $c_i = \sum_j \sigma(s_j - s_i)$; unlike the hard counts, evaluating the soft counts exactly costs $O(n^+ n^-)$ (they can be approximated in near-linear time, e.g. by binning the opposite-class scores, but we do not rely on this).

Proof. By the chain rule, $\nabla_\theta L_{\text{pair}} = \sum_k (\partial L_{\text{pair}}/\partial s_k) \nabla_\theta s_k$, and $\partial L_{\text{pair}}/\partial s_i = \sum_j \ell'(s_i - s_j) = -c_i$ for positives, $\partial L_{\text{pair}}/\partial s_j = -\sum_i \ell'(s_i - s_j) = c_j$ for negatives. The frozen-weight linear objective

Algorithm 1 FW-BPR: iteratively reweighted bipartite ranking

Require: weight-accepting learner $\mathcal{A}$, data $\{(x_k, t_k)\}_{k=1}^{N}$, margin $m \geq 0$, rounds T

$c_k \leftarrow n^-$ for positives, $c_k \leftarrow n^+$ for negatives {balanced start = first FW step}
for $t = 0, 1, \ldots, T-1$ **do**
  $f_t \leftarrow \mathcal{A}\big(\{(x_k, t_k)\}$, weights $\propto c\big)$; $s_k \leftarrow f_t(x_k)$
  $c_i \leftarrow \#\{j : s_i - s_j < m\}$ for positives; $c_j \leftarrow \#\{i : s_i - s_j < m\}$ for negatives $\{O(N \log N)\}$
  if $\sum_k c_k = 0$ **or** counts unchanged **then**
    break
  end if
end for
return $f_{t^\star}$, $t^\star$ chosen on validation data (cross-fitted)

has the same per-score derivatives, hence the same θ-gradient. The hinge and logistic weight formulas are immediate; the $O(N \log N)$ claim follows by sorting scores and locating each $s_i - m$ among opposite-class scores by binary search. □

Proposition 1 says count-reweighted training *is* pairwise training, one gradient step at a time, for any architecture: an SGD trajectory on the pairwise loss coincides with an SGD trajectory on the count-weighted univariate objective whose weights are refreshed at every step. Refreshing every step is free for the hinge ($O(N \log N)$); refreshing less often (per epoch, as in Section 6) gives a lagged approximation. Two practical caveats follow directly from the formula. First, if the model interpolates the training ranking, all hard counts vanish and the update stalls; a positive margin m (or soft counts) keeps the weights alive. Second, the identity concerns *gradients*, not minimizers: with weight decay, normalization layers, or adaptive optimizers, lagged reweighting and true pairwise training may drift apart, and we observe (Section 6) that the drift is within noise at per-epoch refresh rates on the problems tried.

5 A Dual-Free Meta-Algorithm

Theorem 1, Corollary 3, and Proposition 1 license the following meta-algorithm, which never touches dual variables and treats the base learner as a black box that accepts per-example weights.

Three readings of Algorithm 1, in decreasing order of rigor. *(a) Linear SVM base learner, margin* 2*:* each round solves the Frank–Wolfe linear-minimization problem of the pairwise dual exactly (Corollary 3); iterates that maintain a convex combination of the visited vertices inherit standard Frank–Wolfe convergence to the RankSVM solution (Jaggi, 2013; Lacoste-Julien et al., 2013), and Corollaries 1 and 2 provide a computable optimality gap. Solving each round's weighted problem to optimality instead (a trust-region variant over the box $[0, \bar{C}\omega]$, Remark 3) is a *relaxation*: its iterates may leave $\mathcal{T}$, in which case they lower-bound the pairwise optimum and the max-flow test certifies the violation. *(b) Differentiable base learner:* by Proposition 1 the procedure performs lagged gradient descent on the pairwise surrogate. *(c) Arbitrary base learner (trees, boosting):* the procedure is a heuristic guided by the same weights; if the learner interpolates the training ranking, the counts vanish and the method provably returns the balanced fit – a failure mode we exhibit empirically below.

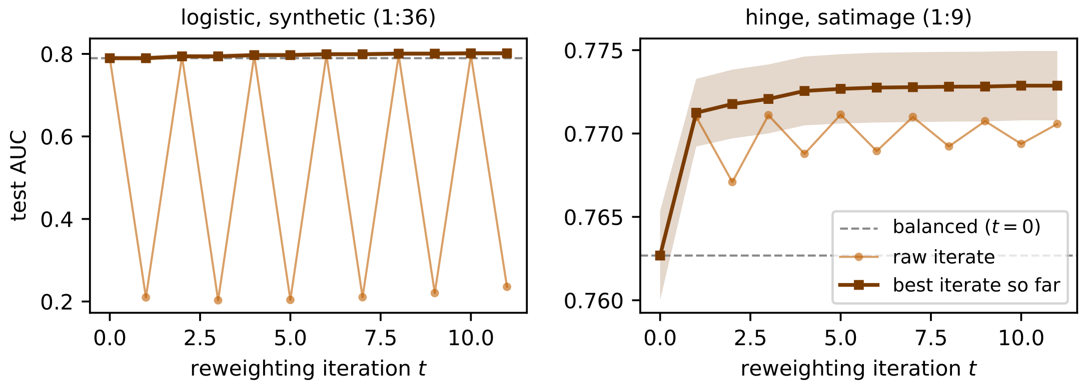

Figure 2: Test AUC by reweighting round (mean $\pm$ standard error over 15 folds, fixed $C = 1$; grey dashed line: balanced fit, $t = 0$). Thin curve: raw iterates. On the extreme-imbalance synthetic set (left, logistic base learner) the raw sequence *oscillates*: a full refit at the new counts is a jump to a polytope vertex with no line search, weight mass concentrates on the few violated examples, and successive vertices over-correct, alternating between strong and inverted rankings. Bold curve: best-iterate-so-far per fold, the quantity harvested by cross-fitted selection (+0.6 points over balanced on the left; right: hinge base learner on satimage). Near-balanced datasets are flat at the dashed line (not shown).

6 Experiments

All code and raw results accompany the paper.[1] We use seven binary datasets spanning imbalance ratios 1:1.5 to 1:43 (breast_cancer, pima, german credit, spambase, satimage class-4, abalone rings ≥ 19, and a synthetic 1:36 Gaussian mixture), with 3-fold $\times$ 5-repeat stratified cross-validation; regularization and the FW-BPR iterate are selected on validation data only. Full tables appear in the supplementary results files; we summarize the four findings.

1. Count reweighting reproduces pairwise training. A linear SVM wrapped in Algorithm 1 (margin 2) matches an explicit RankSVM (pairwise hinge on difference vectors) on all seven datasets: mean paired AUC differences are at most 0.004 and per-fold wins split evenly. The same holds for MLPs: per-epoch count-refreshed weighted cross-entropy tracks full pairwise-logistic training within noise on all four datasets tried (winning 5/5 and 4/5 paired splits on two of them), while a pairwise epoch costs 1.3–4.4$\times$ more at these sizes and scales as $O(n^+n^-)$ against $O(N \log N)$. The gradient identity itself was verified through an MLP by automatic differentiation: hinge gradients agree *exactly* (difference 0 in float64), logistic gradients to 10^{-13}.

2. Iterating is a free option over balancing. With cross-fitted joint selection of (C, t), FW-BPR-logistic gains +0.6 AUC points over balanced logistic on the dataset with headroom (synthetic 1:36, winning 10/15 folds, median selected $t = 6$) and *selects* $t = 0$ – i.e., returns the balanced solution, at no loss – where balancing is already optimal (Figure 2). Selection

[1] Verification suite and experiment harness in the project repository; the theorem suite runs 241 numerical checks of every claim in Section 3 (duality gaps $\leq 10^{-10}$, Gale/max-flow agreement on all sampled points).

over iterates is not optional: the raw refit sequence is vertex-hopping without line search and can oscillate on extreme imbalance (Figure 2, left) – the discrete analogue of Frank–Wolfe's need for step sizes or convex combinations (Jaggi, 2013). Under naive single-split selection the harvestable gains are destroyed by selection noise; the small validation sets of heavily imbalanced problems contain few positives, and this, not the optimization, is the binding constraint.

3. The negative results are the theory's own predictions. Balanced logistic regression already matches RankSVM everywhere (winning up to 14/15 folds on some datasets) – the first Frank–Wolfe step plus AUC-consistency (Kotłowski et al., 2011; Agarwal, 2014) predicts precisely this – and neither pairwise training nor its count-reweighted equivalent broadly beats plain or balanced cross-entropy on generic tabular data. Any contrary claim would contradict the consistency of the univariate baselines. The practical role of the equivalence is therefore cost reduction and certification wherever pairwise objectives are genuinely the right tool (ranking heads, verification, deep AUC maximization (Yuan et al., 2021)), not headline accuracy.

4. Interpolating learners collapse, as predicted. Wrapping gradient-boosted trees drives training AUC to 1.0, all hard counts to zero, and Algorithm 1 provably returns the balanced fit (identical AUC to four decimals); static balancing itself does not help boosting on these datasets. Rescuing interpolating learners requires out-of-fold count estimation or calibrated margins, which we leave open.

7 Discussion

The pairwise hinge for bipartite ranking and balanced cost-sensitive classification are one optimization problem seen from two coordinate systems, glued by a transportation polytope. The efficiency tricks of Joachims (2005) and Chapelle and Keerthi (2010), the lambdas of LambdaMART, and the hardness weights of example-mining heuristics are all shadows of the same object; the polytope adds what the tricks lack – an exact domain, a membership certificate, and a sandwich bound on the remaining gap. The equivalence also delimits its own practical value: since balanced univariate training is the first step of the pairwise optimization and is already AUC-consistent, no method in this family can dominate strong univariate baselines at scale, and our experiments bear this out. Where pairwise objectives earn their cost – verification, metric-learning decision layers, extreme-imbalance regimes with limited positives (Yang and Ying, 2022) – the identity of Proposition 1 replaces them at univariate cost, and the certificate says how far from pairwise-optimal any reweighted solution sits. Extending the count refresh to interpolating learners via cross-fitting, and testing the substitution inside deep AUC-maximization pipelines, are the natural next steps.

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